Tanmay Pandey

Integrated BS–MS Student
Indian Institute of Science Education and Research (IISER) Mohali

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Research Profile

I am a biophysics student at IISER Mohali working on membrane mechanics, plasma membrane vesicles, and quantitative image analysis. My research focuses on studying membrane morphology and mechanical properties using fluorescence microscopy and computational analysis tools. My current work involves quantitative analysis of giant plasma membrane vesicles (GPMVs) and biomimetic membrane systems.

Education

2022–2027 **BS–MS Dual Degree in Biology (Physics Minor)**
Indian Institute of Science Education and Research (IISER) Mohali

Publications and Research Outputs

2026 **Directed evolution effectively selects for DNA-based physical reservoir computing networks**
T. Pandey, P. Feketa, J. Steinkühler
Neuromorphic Computing and Engineering (IOP).
DOI: 10.1088/2634-4386/ae2cc3

2025 **Experimentally determined shapes of plasma membrane vesicles and PC/PC–cholesterol vesicles**
H. Kaur, R. Kudawla, **T. Pandey**, T. Bhatia
Journal of Physical Chemistry B (ACS).
DOI: 10.1021/acs.jpcb.4c07431

2025 **Vesicle Shape Analysis Toolkit**
T. Pandey, and T. Bhatia
Open-source software for contour extraction, curvature, and mechanical analysis of vesicle membranes (Zenodo + GitHub)
DOI: 10.5281/zenodo.17104430

2024 **Shape analysis of biomimetic and plasma membrane vesicles**
R. Kudawla, H. Kaur, **T. Pandey**, T. Bhatia
ChemSystemsChem (Wiley).
DOI: 10.1002/syst.202400052

Research Experience

2023–2026 **Membrane Mechanics and Plasma Membrane Vesicles**
Project Intern, Soft Matter Biophysics Lab, IISER Mohali
Supervisor: Dr. Tripta Bhatia

- Quantitative analysis of GPMV and GUV morphology to extract curvature, reduced volume, and membrane mechanical parameters.
- Developed Python-based pipelines for contour detection, fluctuation analysis, and vesicle classification.
- Performed GUV electroformation, GPMV preparation, and multi-channel fluorescence microscopy.

2024–2025 **DNA-Based Physical Reservoir Computing Systems**
Student Assistant, Bio-inspired Computation Lab, University of Kiel.
Remote Internship
Supervisor: Prof. Jan Steinkühler

- Developed numerical models of DNA-based spring–mass systems as physical reservoirs inspired by intracellular polymer dynamics.
- Analyzed adaptability and performance of physical reservoir networks across nonlinear dynamical tasks.

Expertise

Microscopy	Fluorescence and confocal microscopy; time-lapse imaging.
Membrane Systems	GUV electroformation; GPMV isolation; membrane fluctuation assays.
Image Analysis	CNN-based segmentation; contour tracking; curvature and fluctuation analysis.
Computation	Python, MATLAB; statistical analysis; machine learning–assisted image pipelines.
Softwares	ImageJ (Fiji), Chimera, PyMol

Administrative Roles

2025–2026 **General Secretary and Hostel Representative**
Student Representative Council (SRC), IISER Mohali

- Representing student concerns at the institute administration level.